Day 2

# From Genomes to Surveillance Action

Genomic Surveillance of Surgical Site Infections & AMR
08:30–17:00

Course files: github.com/aryakaul/ughe-ssi-genomics-course

# Day 2 · Morning

| Time | Session |
|---|---|
| 08:30 | Recap: your questions from yesterday |
| 08:50 | **Lab 4:** Who is it, and what does it resist? |
| 10:30 | *Break* |
| 10:45 | **Lecture 4:** Typing, trees and outbreaks |
| 11:30 | **Lab 5:** Pathogenwatch and Microreact |
| 12:30 | *Lunch* |

# Day 2 · Afternoon

| Time | Session |
|---|---|
| 13:15 | **Lab 6:** Case study, a cluster on the surgical ward |
| 15:00 | *Break* |
| 15:15 | **Lecture 5:** Building a sustainable programme |
| 16:00 | Action planning |
| 16:40 | Knowledge check, evaluation, certificates |

# Your questions from yesterday

## Answers to the exit tickets

> Today's files: github.com/aryakaul/ughe-ssi-genomics-course
> Lab 5–6 data are in `labs/data/`

Lab 4

# Who Is It, and What Does It Resist?

Day 2 · 08:50–10:30 · 100 min
Pairs · swap driver and navigator at Part D

LO4: detect AMR genes and plasmids, type the isolate, and interpret the results alongside phenotypic AST

# Open your handout

## Lab 4 handout

`Lab4_species_typing_amr.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

# The scenario

- A patient develops a **deep SSI 9 days** after an open fracture repair
- The lab reports *S. aureus*, **methicillin-resistant**
- The isolate was sequenced: you have its **assembly** (FASTA)

**KEY MESSAGE**

**What can the genome add?**

Teaching scenario. The genome is real public data from the Galaxy Training Network.

# Plan for this lab

| Part | What | Time |
|---|---|---|
| A | Get the genome | 5 min |
| B | Who is it? Species and strain type | 20 min |
| C | What does it resist? AMR genes | 35 min |
| D | Is resistance mobile? Plasmids | 15 min |
| E | Does it match the lab’s AST? | 20 min |

# Get the genome

1. New history: `Lab4 – AMR`
2. **Upload** → **Paste/Fetch data** → paste the URL **from your handout**
3. **Type:** `fasta` · **Name:** `MRSA_contigs.fasta`

> **OPTIONAL**
>
> Your own Lab 2 assembly finished? Run every step on it too.

# MLST: the strain type

Tool **MLST** → `MRSA_contigs.fasta` → scheme **automatic** → **Run**

```
MRSA_contigs.fasta   saureus   ST?   arcC(?) aroE(?) …
```

**CHECK YOURSELF**

1. Which **scheme** did MLST choose? What does it tell you?
2. What is the **ST**? Is it a well-known MRSA clone?

PART B2 · OPTIONAL

# Species check with Kraken2

- Tool **Kraken2** on `MRSA_contigs.fasta`, with a **report**
- What is the top species?

> **KEY MESSAGE**
>
> **If the organism is wrong or the culture is mixed, every result after this is wrong.**

# AMRFinderPlus (NCBI)

- **Input type:** nucleotide → `MRSA_contigs.fasta`
- **Organism:** `Staphylococcus_aureus` (finds point mutations too)
- Everything else: defaults → **Run** → open the **report**

**KEY COLUMNS**

**Element symbol** (gene) · **Class** (drug class) · **% Identity** · **% Coverage**

PART C2

# staramr

- ResFinder + PointFinder + PlasmidFinder + MLST in one tool
- Input `MRSA_contigs.fasta` → defaults → **Run**
- Open **summary.tsv** and **detailed_summary.tsv**

PART C3

# Record your findings

| Gene / mutation | Drug class | AMRFinderPlus? | staramr? | On a plasmid? |
|---|---|---|---|---|
| | | | | |
| | | | | |
| | | | | |

Fill in the full table in your handout. The last column comes in Part D.

PART C

# Check yourself: AMR genes

**CHECK YOURSELF**

1. Which gene explains the **methicillin resistance**? *(Glossary cheat sheet!)*
2. Did both tools find the same genes? If not, why might they differ?
3. What is staramr's **predicted phenotype**?

# Swap driver and navigator

The navigator takes the keyboard; the driver reads the handout

# Is resistance mobile? Plasmids

1. In `detailed_summary.tsv`, find the **plasmid** rows (`rep…`) and note their **contig**
2. Mark resistance genes on the **same contig** as a plasmid

> **ALTERNATIVE**
>
> **ABRicate** with the **plasmidfinder** database

PART D

# Check yourself: plasmids

**CHECK YOURSELF**

1. How many plasmid replicons? On which contigs?
2. Which resistance genes are probably on a **plasmid**? Why does that matter for infection control?
3. Why can’t we be **certain** from an Illumina assembly?

# Does the genome match the lab's AST?

- Use the **AST report in your handout** (Part E)
- **It is hypothetical:** written for teaching, not the real AST for this genome
- For each antibiotic: which gene explains it? Do genome and AST **agree**?

PART E

# Check yourself: genome vs AST

**CHECK YOURSELF**

1. Where do genome and AST **disagree**? Give two possible explanations for each.
2. What would you **tell the clinician**? Write one plain-language sentence.
3. Would you use the genome **instead of** AST to treat this patient?

# What each tool told us

| Tool | Answers |
|---|---|
| **MLST** | **Who?** Strain type, known clones |
| **AMRFinderPlus / staramr** | **What resistance?** Genes and mutations |
| **PlasmidFinder** | **Is it mobile?** |
| **Comparing with AST** | **How confident are we?** |

Next: what about **many** isolates? → Lecture 4 and Lab 5

# Extension for fast pairs

- **ABRicate** with the **card** and **vfdb** (virulence) databases
- Annotate the genome with **Bakta**
- Upload the FASTA to **ResFinder** on the web and compare

Adapted from the GTN tutorial “Identification of AMR genes in an assembled bacterial genome” (CC-BY 4.0).

# Break

Back at 10:45 for Lecture 4

Lecture 4

# Typing, Trees and Outbreaks

Day 2 · 10:45–11:30 · 45 min

LO5: type isolates and interpret a phylogeny to support or rule out an outbreak

**“We’ve had 5 *Klebsiella* SSIs on Ward B in 6 weeks. Is this an outbreak?”**

# What is an outbreak?

- An outbreak means **transmission**: the same strain passing between patients
- Directly, or through staff, equipment or the environment

**KEY MESSAGE**

**Same species ≠ same strain.**
**Same AST pattern ≠ same strain.**

# Typing: naming bacteria at different resolutions

| Method | Like a… | Used for |
|---|---|---|
| Species | country | Clinical ID |
| **MLST** (7 genes) → ST | city | Is it a known high-risk clone? |
| **cgMLST** (1,000–3,000 genes) | street | Clusters within an ST |
| **SNPs** (whole genome) | house number | Outbreaks and transmission |

# MLST in one slide

- Seven “housekeeping” genes; each version gets a number
- The combination of 7 numbers = the **Sequence Type (ST)**
- Known high-risk clones: *K. pneumoniae* ST258, ST307 · MRSA ST8, ST22
- Tools do it automatically (Lab 4, Lab 5)

**KEY MESSAGE**

**Same ST = possibly related. Different ST = not a recent outbreak.**

# SNP distances

- Align genomes to a reference
- Count the positions where they differ
- **Few SNPs** → recent common ancestor → possible transmission
- **Many SNPs** → not recent transmission

**ROUGH GUIDE (VARIES BY STUDY)**

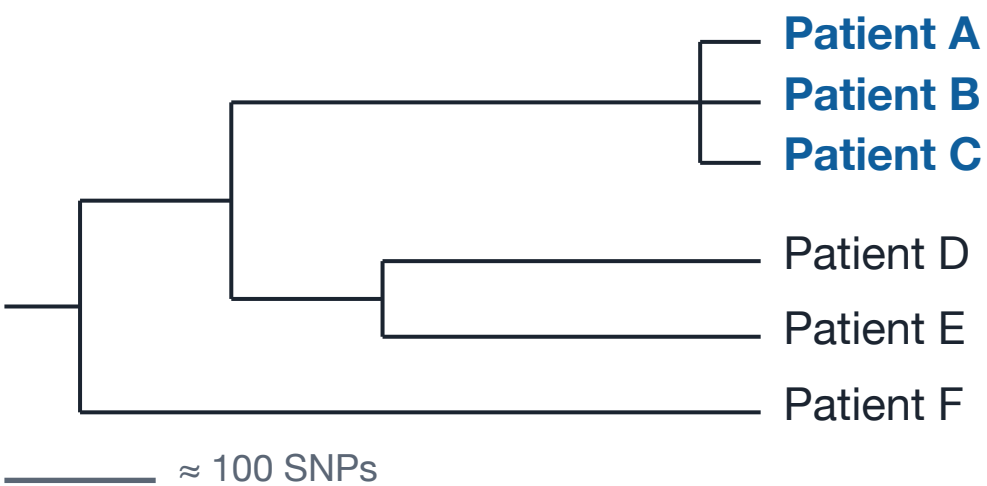
*S. aureus*: **≤ 15–30 SNPs**

*K. pneumoniae*, *E. coli*: **≤ 10–25 SNPs**

**KEY MESSAGE**

**Thresholds are guides, not laws. Always combine them with epidemiology.**

# Reading a phylogenetic tree




- **Tips** = isolates
- **Branch length** = genetic change
- Tips on **short branches** close together are closely related (A, B, C)

**COMMON MISTAKES**

- Reading top-to-bottom order as meaningful
- Ignoring the **scale bar**

# Genomics + epidemiology: time, place, person

A genomic cluster is an outbreak only if the **epidemiology fits**: same ward or theatre at overlapping times? Same surgeon, equipment or staff?

**Confirm**
Same strain, few SNPs, overlapping stays

**Refute**
Different STs: no need to close the ward

**Find hidden links**
Cases on different wards linked by shared equipment or staff

# Strain outbreak vs plasmid outbreak

**STRAIN OUTBREAK**

The **same clone** in many patients

Tree: one tight cluster

**PLASMID OUTBREAK**

**Different** strains or species, but the **same resistance plasmid**

Needs Nanopore or hybrid data to confirm

**KEY MESSAGE**

**Tree says “unrelated” but the resistance is unusual and identical? Think plasmid.**

# Vote: outbreak or not? (cards A / B / C)

1. **4 MRSA**, all ST22, 2–5 SNPs apart, same ward over 3 weeks
2. **4 *Klebsiella***: ST15, ST307, ST14, ST101
3. **3 *E. coli***, all ST131, 80–150 SNPs apart, 3 different hospitals

# Vote: answers

1. **Likely outbreak**
2. **Not a single-strain outbreak.** Check for a shared plasmid?
3. **A common circulating clone**, not recent transmission

# From result to action

| Finding | Possible IPC action |
|---|---|
| Confirmed cluster | Enhanced cleaning, cohorting, contact screening, theatre audit, staff screening if indicated |
| Refuted cluster | Avoid unnecessary ward closure; look at other causes (e.g. prophylaxis timing) |
| High-risk clone or new carbapenemase | Notify the national reference lab / public-health authority |

> Communicate in **plain language** to clinicians and managers. You'll practise this in Lab 6.

# Tools for the next lab

**PATHOGENWATCH**

Upload assemblies → species, ST, AMR, clustering, tree

**MICROREACT**

Tree + map + timeline + metadata in one interactive view

Both are free, used by national public-health labs, and need **no coding**.

Lab 5

# Pathogenwatch and Microreact: Many Genomes at Once

Day 2 · 11:30–12:30 · 60 min
Pairs

LO5: use free web platforms to type, screen and visualise a collection of genomes

# Open your handout

## Lab 5 handout

`Lab5_pathogenwatch_microreact.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

# The scenario

- The IPC nurse at **District Hospital A** is worried about more **MRSA SSIs on Surgical Ward B**
- The lab stored the isolates and sent them for sequencing
- You have **genomes from all MRSA isolates of the past months**

**KEY MESSAGE**

**Get a quick overview of all the genomes at once.**

The genomes are real public data. The hospital, wards and dates are fictional.

# Plan for this lab

| Part | What | Time |
|---|---|---|
| A | Pathogenwatch: typing and AMR | 30 min |
| B | Microreact: tree + timeline | 25 min |
| | Summary | 5 min |

# Files from the instructor

| File | Contains |
|---|---|
| `assemblies/` | One FASTA per isolate |
| `koser_tree.nwk` | Phylogenetic tree (Galaxy: Snippy → IQ-TREE) |
| `lab6_metadata.csv` | Patient, ward, procedure and date information |

- **GitHub:** github.com/aryakaul/ughe-ssi-genomics-course
- Green **Code** button → **Download ZIP** → unzip → `labs/data/`

# Upload and explore

1. Log in to `pathogen.watch` → **Upload** → drag in the FASTA assemblies
2. Wait for each genome to be analysed (a few minutes)
3. For **at least 6** genomes, record:

| Genome | Species | ST | Predicted resistances | QC pass? |
|---|---|---|---|---|
| | | | | |
| | | | | |

Does the organism agree with the lab? Check the assembly QC metrics (length, contigs, N50).

# Make a collection and view the tree

1. Select all your genomes → **Create collection** → name it `UGHE Lab 5`
2. Open the collection → explore the **tree** view
3. Explore the **AMR** view: resistance profiles side by side

**CHECK YOURSELF**

1. How many different STs are in the collection?
2. Do all isolates have the **same** AMR profile? What does that suggest, and what does it **not** prove?
3. In the tree, can you see a group of isolates on very **short** branches?

# Create a project

1. Go to `microreact.org` → **Upload** (sign in only if you want to save)
2. Drag in **both** files: `lab6_metadata.csv` and `koser_tree.nwk`
3. Column that links the table to the tree tips → **`id`**
4. Dates: confirm that **`year`**, **`month`**, **`day`** form the date
5. **Continue / Create**

> “No matching IDs”? Raise your hand: the tree labels and the `id` column must match exactly.

# Explore: tree + time + place

- **Colour** the tree by `ward`, then `theatre`, then `hospital`
- Open the **Timeline**: when did each infection happen?
- **Click a tip**: its row lights up in the table and timeline
- **Lasso-select** a group of close tips: where and when are they?

**CHECK YOURSELF**

1. Is there a group that is closely related **and** close in time **and** in the same place?
2. Is any isolate from the **same ward at the same time not** in that group?
3. Is any isolate in the group from a **different ward**? What could link it?

PART B3

# Save, share, summarise

**Save** (if signed in) and copy the project link. **You'll use it in Lab 6.**

| Tool | Gives you | Input |
|---|---|---|
| **Pathogenwatch** | Species, ST, AMR, QC, clustering, global context | FASTA assemblies |
| **Microreact** | Visual story: tree + time + place + metadata | Tree (Newick) + table (CSV) |

Both are free and used by national public-health labs. **You can use them on Monday with your own data.**

Pathogenwatch: Centre for Genomic Pathogen Surveillance. Microreact: Argimón S, et al. Microb Genom 2016.

# Lunch

Back at 13:15 for the case study (Lab 6)

Lab 6

# Case Study:
# A Cluster on the Surgical Ward

Day 2 · 13:15–15:00 · 105 min
Groups of 4

LO3–LO5: combine genomic and epidemiological evidence to decide whether an outbreak is happening, and communicate the decision

# Open your handout

## Lab 6 handout

`Lab6_ssi_outbreak_case_study.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

# About this case

- The **genomes are real**: MRSA isolates from a published outbreak investigation
- The **hospitals, wards, patients and dates are fictional**, so the case feels like our setting

> The real story is revealed at the end.

THE SITUATION

# District Hospital A · Monday, 16 March 2026

The IPC nurse writes to you:

> *“Since early February we’ve had **more MRSA wound infections than usual on Surgical Ward B**. The surgeons say it’s bad luck. Theatre staff say it’s the ward. The ward says it’s the theatre.*

# The question

> *The director asks whether we should **close Ward B**. That would cancel about 40 operations a week. We got genomes back last week for all MRSA isolates since September.*
>
> ***Can you tell us what's going on by Wednesday's IPC meeting?"***

# Your evidence

1. **Microreact project** from Lab 5: tree + timeline + metadata
2. **SNP matrix:** `koser_snp_matrix.tsv` (opens in Excel)
3. **Pathogenwatch results** from Lab 5: ST and AMR
4. **Metadata:** `lab6_metadata.csv`

# Group roles

| Role | Job |
|---|---|
| **Genomics lead** | Reads the tree and SNP matrix |
| **Epidemiologist** | Builds the timeline: who was where, when |
| **Clinician / IPC** | Thinks through transmission routes and actions |
| **Communicator** | Writes and gives the 5-minute briefing |

# Plan for this lab

| Part | What | Time |
|---|---|---|
| A | Describe the cases | 15 min |
| B | What does the genome say? | 25 min |
| C | Genomics + epidemiology together | 20 min |
| D | Prepare the briefing | 25 min |
| E | Briefings to the IPC committee | 15 min |

# Describe the cases

Sketch an **epidemic curve**: weeks across, number of cases up.
Two colours: **Ward B** and **other wards / hospitals**.

**QUESTIONS**

1. How many MRSA isolates in total? How many from Ward B since 1 February?
2. Do all Ward B cases look the same on the **lab report**? Does that make them one outbreak?

# What does the genome say?

1. **Sequence types:** group the isolates by ST (Pathogenwatch)
2. **SNP distances:** highlight all pairs **≤ 30 SNPs** apart
3. Which isolates form a **cluster**, all close to each other?
4. **Tree:** colour by **ward**, then by **theatre**. Is the cluster a tight group on short branches?

≤ 30 SNPs: our working threshold for possible recent *S. aureus* transmission. Published thresholds vary.

# Put genomics and epidemiology together

For **each** Ward B case since February:

| Patient | In the cluster? | Ward / theatre / time link? | Linked · not linked · uncertain |
|---|---|---|---|
| | | | |
| | | | |

Use the full table in your handout.

PART C

# Discuss in your group

1. Is there an outbreak? How confident are you?
2. Is a Ward B patient **outside** the cluster, though on the ward at the same time?
3. Is a cluster patient **not** on Ward B? What do they share? → **likely route?**
4. Are isolates from **other hospitals** in the cluster?
5. What **other information** would you want?

# Prepare a 5-minute briefing

For the IPC committee: director, chief surgeon, nursing lead, lab head. **Not genomics experts.**

**RULES**

- One flip-chart page or 3 slides
- Explain any jargon: “SNP” → “genetic differences”
- **No more than one figure**

PART D

# Briefing template

1. **Bottom line:** “We found / did not find evidence of an outbreak of ___ linked to ____.”
2. **What we did:** “We sequenced __ MRSA isolates from __ to __.”
3. **What we found:** one visual + 2–3 bullets
4. **What it means:** which cases are linked; the likely route
5. **Recommendations:** 3 actions, with who and by when
6. **What we don’t know yet**

# Briefings to the IPC committee

Each group has **5 minutes**. The committee must ask at least one question:

**“Should we close Ward B?”**

**“Is it the surgeons' fault?”**

**“How sure are you?”**

**“What will this cost?”**

# Pause

Instructor: don’t go to the next slide until every group has given its briefing

# The reveal

The genomes come from **Köser et al., *NEJM* 2012**: a suspected MRSA outbreak in a **neonatal unit** in Cambridge, UK.

- Sequencing separated the **outbreak isolates** from unrelated MRSA that looked identical on routine tests
- A follow-up study (**Harris et al. 2013**) traced transmission into the community and found a **staff carrier**, which ended the outbreak

Genomic data: ENA project PRJEB2912 (Köser et al. 2012). Scenario metadata are fictional and created for teaching.

# Take-home messages

1. **Same species + same antibiogram ≠ same outbreak**
2. Genomics finds **hidden links**: shared theatre, staff, equipment
3. Genomics can **prevent unnecessary actions**, like closing a ward
4. Genomics is only useful with **good metadata** and **clear communication**

# Break

Back at 15:15 for Lecture 5

Lecture 5

# Building a Sustainable Genomic Surveillance Programme

Day 2 · 15:15–16:00 · 45 min
≈ 25 min talk + 20 min discussion

LO6: design a sustainable, outsourced genomic surveillance workflow for your own setting

# What we can do on Monday

**Collect** → **Store** → **Send** → **Analyse** → **Act**

No sequencer is needed to start. We need:

- a **plan** and a clear question
- **stored isolates** with good **metadata**
- a sequencing **partner**
- **people** who can analyse the data

# Choose a question first

| Type | Example | What to sequence |
|---|---|---|
| **Outbreak** (reactive) | “Is this ward cluster one strain?” | All cluster isolates + a few unrelated controls |
| **Routine** (proactive) | “Which ESBL/ carbapenemase genes cause our SSIs?” | e.g. all resistant SSI isolates for 6 months |
| **Research** | “Are post-caesarean SSI *E. coli* from the community?” | A designed sample with matched community isolates |

# Start small.

50–100 well-chosen isolates with complete metadata beat 1,000 random ones.

# Sampling and biobanking

- Store **every** SSI isolate with resistance of interest (at least ESBL, carbapenem-resistant, MRSA)
- –80 °C freezer or glycerol stocks; a **backup** at a second site if possible
- Keep an **isolate register** (spreadsheet): study ID, organism, date, ward, specimen, AST, freezer location
- Agree on labelling **before** you start

# Metadata and data management

- Use a template aligned with **PHA4GE** fields
- **Coded IDs only.** Keep the ID → patient key secure and separate
- Name files consistently: `UGHE-SSI-0001_R1.fastq.gz`
- Back up raw reads in **two places**: they can't be replaced
- Galaxy histories are a free, reproducible record of every analysis

# Ethics, agreements and data sharing

- **Surveillance and IPC:** usually covered by routine public-health practice
- **Research:** needs ethics approval (Rwanda National Ethics Committee + institutional IRB)
- **Sending isolates abroad:** material transfer agreement (MTA)
- **Data-sharing agreement:** who owns the data and how it is published
- **Share genomes publicly** (ENA / NCBI) with non-identifying metadata

**KEY MESSAGE**

**Equity: local investigators should lead the analysis and authorship.**

# Partners in the region

**RBC**
Rwanda Biomedical Centre / National Reference Laboratory

**Africa CDC PGI**
Regional sequencing and training network

**ASLM**
AMR surveillance programmes such as MAAP

**Other partners**
Universities, NGOs, Fleming Fund projects, commercial services

**Ask the room:** which of these do we already work with?

# Reporting results so they change practice

| Audience | What they need |
|---|---|
| **Clinicians** | “This isolate carries an NDM carbapenemase. Avoid carbapenems; consult ID.” |
| **IPC committee** | Cluster summary, tree, timeline, recommended actions |
| **Ministry / RBC** | Quarterly summary of resistance genes and clones |

> Use simple visuals (Microreact) and a one-page template. **Close the loop:** did the action reduce SSIs?

# People: keeping the skills alive

- **2–3 “genomics champions”** per department
- A monthly **data clinic / journal club** for course graduates
- Self-paced learning: Galaxy Training Network, The Carpentries, Africa CDC, Wellcome Connecting Science
- Build genomics into **student teaching** (MGHD, MBBS research projects)

# Funding

- Build sequencing costs into grants (AMR, surgery, maternal health)
- Funders active in Africa: Fleming Fund, Wellcome, Gates Foundation, Africa CDC, NIH / Fogarty, EDCTP

**BUDGET PER ISOLATE**

Sequencing + shipping + staff time + data storage

# Your one-page action plan

1. **Question:** what do I want to know?
2. **Isolates:** which, how many, from where, when? Already stored?
3. **Metadata:** which fields? Who collects them?
4. **Partner:** who? Which platform? Cost? Timeline?
5. **Analysis:** which tools? Who does it?
6. **Ethics:** IRB? MTA? Data sharing?
7. **Communication:** who gets results, in what format?
8. **First step next week:** _____

# Discussion

20 min · small groups, then the whole room

1. What is the **single biggest barrier** to starting genomic surveillance at our institution? How could we overcome it?
2. Which **question** (outbreak, routine, research) would be most valuable to answer first?

# Action planning

Draft **your own one-page plan** for a genomic surveillance mini-project.

- Use the 8-point template from Lecture 5
- Choose a question you could really start on next week
- Be concrete: **which** isolates, **which** partner, **who** does what

**KEY MESSAGE**

**Last point on the template: your first step next week.**

# Knowledge check: after the course

- The **same questions** as yesterday
- Use the **same code word** as yesterday, and write **Post**
- Afterwards we'll go through the answers together

# Knowledge check · Post

- **Not graded**, and anonymous
- On your quiz sheet or a piece of paper, write your **code word** and **Post**
- For each question, write its number and your answer (e.g. “1 b”)
- One question at a time on the screen

**KEY MESSAGE**

**It’s fine not to know an answer. Just leave it blank.**

KNOWLEDGE CHECK · QUESTION 1 OF 15

# Which question can whole-genome sequencing answer that standard culture + AST cannot?

a) Which antibiotic should this patient receive today

b) Whether isolates from several patients are the same strain

c) Whether the organism is Gram-positive

d) Whether the patient has a fever

KNOWLEDGE CHECK · QUESTION 2 OF 15

# Do you need to own a sequencer to do genomic surveillance? (Yes / No) Briefly explain why.

Write your answer on your paper

KNOWLEDGE CHECK · QUESTION 3 OF 15

# Name two bacterial species that commonly cause surgical site infections.

Write your answer on your paper

KNOWLEDGE CHECK · QUESTION 4 OF 15

# An Illumina sequencing run typically produces reads that are:

a) 100–300 bases long and highly accurate

b) 10,000+ bases long

c) the whole genome in one read

d) protein sequences

# Which technology is better at showing whether a resistance gene is on a plasmid?

a) Illumina short reads alone

b) Oxford Nanopore long reads

c) AST

d) MLST

KNOWLEDGE CHECK · QUESTION 6 OF 15

# A genome is 5 million bases. You have 1 million reads of 150 bases. What is the approximate coverage?

a) 3×

b) 30×

c) 150×

d) 300×

# What does a FASTQ file contain that a FASTA file does not?

a) Sequence names

b) DNA letters

c) Quality scores

d) Gene names

# After assembly, which result usually means a *better* assembly?

a) More contigs and a smaller N50

b) Fewer contigs and a larger N50

c) A total length twice the expected genome size

d) No contigs

# The gene *mecA* in *S. aureus* indicates:

a) Vancomycin resistance

b) MRSA (methicillin resistance)

c) Carbapenem resistance

d) A virulence toxin

# *blaNDM-1* is an example of:

a) An ESBL

b) A carbapenemase

c) An efflux pump

d) A housekeeping gene

# The genome shows no known resistance gene, but AST says “resistant”. Give one possible explanation.

Write your answer on your paper

# Two *Klebsiella* isolates are ST15 and ST307. Could they be part of the same recent transmission chain?

a) Yes, very likely

b) No, they are different strain types

c) Only if they are both ESBL

# On a phylogenetic tree, what tells you how genetically different two isolates are?

a) Their order from top to bottom

b) The branch lengths between them

c) The colour of the labels

# Four isolates are 2–6 SNPs apart and the patients overlapped on the same ward. This most strongly suggests:

a) Unrelated infections

b) Possible transmission / outbreak

c) Contamination of the sequencer

KNOWLEDGE CHECK · QUESTION 15 OF 15

# List three pieces of metadata that must be recorded with every isolate for surveillance.

Write your answer on your paper

KNOWLEDGE CHECK

# How confident are you?

Rate each from **1** (not at all) to **5** (very confident):

- I can explain what a genome sequence can tell us about an SSI
- I can run a tool in Galaxy
- I can identify resistance genes from a genome
- I can interpret a phylogenetic tree
- I can plan a genomic surveillance project

# N4

## What further training or support would help you most after this course?

# Let’s go through the answers

Hand in your paper first, then follow along

# Which question can whole-genome sequencing answer that standard culture + AST cannot?

a) Which antibiotic should this patient receive today

**b) Whether isolates from several patients are the same strain** ✓

c) Whether the organism is Gram-positive

d) Whether the patient has a fever

**WHY**

Culture and AST guide treatment for this patient. Only the genome shows whether isolates are the **same strain**.

# Do you need to own a sequencer to do genomic surveillance? (Yes / No) Briefly explain why.

**ANSWER**

**No.** Isolates or DNA can be sent to a partner. The local skills are choosing isolates, recording metadata, analysing and acting on results.

# Name two bacterial species that commonly cause surgical site infections.

**ANSWER**

Any two of: *S. aureus*, *E. coli*, *K. pneumoniae*, *P. aeruginosa*, *A. baumannii*, *Enterobacter*, *Enterococcus*.

# An Illumina sequencing run typically produces reads that are:

**a) 100–300 bases long and highly accurate** ✓

b) 10,000+ bases long

c) the whole genome in one read

d) protein sequences

**WHY**

Illumina gives short, very accurate reads. Long reads come from Nanopore.

# Which technology is better at showing whether a resistance gene is on a plasmid?

a) Illumina short reads alone

**b) Oxford Nanopore long reads** ✓

c) AST

d) MLST

**WHY**

Long reads can span a whole plasmid, so you can see which genes sit on it.

# A genome is 5 million bases. You have 1 million reads of 150 bases. What is the approximate coverage?

a) 3×

**b) 30×** ✓

c) 150×

d) 300×

**WHY**

1,000,000 × 150 ÷ 5,000,000 = **30×**

# What does a FASTQ file contain that a FASTA file does not?

a) Sequence names

b) DNA letters

**c) Quality scores** ✓

d) Gene names

**WHY**

FASTQ = sequence + a **quality** line for every read. FASTA = sequence only.

# After assembly, which result usually means a *better* assembly?

a) More contigs and a smaller N50

**b) Fewer contigs and a larger N50** ✓

c) A total length twice the expected genome size

d) No contigs

**WHY**

Fewer, longer pieces. A total length twice the expected size suggests contamination.

# The gene *mecA* in *S. aureus* indicates:

a) Vancomycin resistance

**b) MRSA (methicillin resistance)** ✓

c) Carbapenem resistance

d) A virulence toxin

**WHY**

*mecA* makes an altered target (PBP2a), so almost all β-lactams stop working.

# `blaNDM-1` is an example of:

a) An ESBL

**b) A carbapenemase** ✓

c) An efflux pump

d) A housekeeping gene

**WHY**

NDM = New Delhi metallo-β-lactamase: it destroys carbapenems.

# The genome shows no known resistance gene, but AST says “resistant”. Give one possible explanation.

**ANSWER**

Any of: an unknown mechanism · a mutation not in the database · low coverage missed the gene · mixed culture · lab error.

# Two *Klebsiella* isolates are ST15 and ST307. Could they be part of the same recent transmission chain?

a) Yes, very likely

**b) No, they are different strain types** ✓

c) Only if they are both ESBL

**WHY**

Different STs are not one recent transmission chain. (They could still share a plasmid.)

# On a phylogenetic tree, what tells you how genetically different two isolates are?

a) Their order from top to bottom

**b) The branch lengths between them** ✓

c) The colour of the labels

**WHY**

Branch length = genetic change. Top-to-bottom order means nothing: branches can rotate.

# Four isolates are 2–6 SNPs apart and the patients overlapped on the same ward. This most strongly suggests:

a) Unrelated infections

**b) Possible transmission / outbreak** ✓

c) Contamination of the sequencer

**WHY**

Very few SNPs **and** an epidemiological link: possible transmission.

# List three pieces of metadata that must be recorded with every isolate for surveillance.

**ANSWER**

Any three of: collection date · facility · ward · specimen type · surgery type and date · AST results · organism · coded patient ID.

# Course evaluation

- What worked well?
- What should we change for the next group?
- Anything still unclear?

**KEY MESSAGE**

**Your answers shape the next version of this course.**

# Keep going

- **Course files:** github.com/aryakaul/ughe-ssi-genomics-course
- Your **Galaxy histories** stay in your account: re-run them on new data
- Next steps: `resources/further_learning.md`
- A monthly **data clinic / journal club** for course graduates

# Thank you!

Certificates now